Practice workbook · Hexadecimal: why we use it and how to convert

Why programmers use hexadecimal (base 16, one hex digit per nibble, shorter and less error-prone than binary) and converting between hexadecimal, binary and denary.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-04

1. Question 3 · part a.i · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

P 7 8 6 3 7 A 0 7 2 0

1. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b only.

2 Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

(2)

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

(2)

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

(1)

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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2. Question 2 · part b · 1 mark

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Further practice · Attempt part b only.

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(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

(1)

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

(1)

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

(2)

(Total for Question 2 = 9 marks)

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2. Question 2 · part b · 1 mark · Working and answer

Working and answer

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3. Question 1 · part a · 1 mark

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Further practice · Attempt part a only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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3. Question 1 · part a · 1 mark · Working and answer

Working and answer

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4. Question 1 · part b · 2 marks

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4. Question 1 · part b · 2 marks · Working and answer

Working and answer

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5. Question 1 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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5. Question 1 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

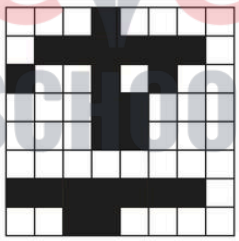


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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5. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a · 2 marks

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Further practice · Attempt part a only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

P 7 2 9 3 7 A 0 5 2 4

6. Question 2 · part a · 2 marks · Working and answer

Working and answer

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7. Question 3 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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Working and answer

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